

Valence Alignment Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Contrasts on the Exchange-Aligned Data Model

PENPAL Project

2026-04-26

Contents

1	Load Data	2
2	Preprocessing	2
3	Descriptive Checks	2
4	Primary: Per-Story Asymmetry Magnitude	3
5	Supplemental: Within-Condition Directional Asymmetry	4
6	Supplemental: Cross-Condition Directional Model	5
7	Supplemental: Directional Scatter Visualizations	7
8	Summary	7

This notebook compares **valence alignment** across the three experimental conditions using the new exchange-aligned structure.

- **Human-AI**: Human co-writing with an LLM
- **Human-Human**: Two humans co-writing together
- **AI-AI**: Two AI agents co-writing together

Important structural note:

- **author_1** / **author_2** are now treated as **slot 1** / **slot 2** in the cleaned exchange sequence.
- In **Human-AI**, those slots also map onto human vs AI roles.
- In **Human-Human** and **AI-AI**, the slots are positional sides of the exchange, not stable identity labels.

The analysis window is restricted to **complete exchanges** and the default aligned turn range (**analysis_turn** 1 to 9).

Primary cross-condition object. Each story contributes one unsigned, label-invariant dyad asymmetry magnitude:

$$A_{\text{valence}}(i) = |\text{atanh}(r_{1 \rightarrow 2, i}) - \text{atanh}(r_{2 \rightarrow 1, i})|$$

where $r_{1 \rightarrow 2, i}$ is the per-story Pearson correlation between same-turn slot 1 and slot 2 valence, and $r_{2 \rightarrow 1, i}$ is the correlation between slot 2's previous-turn valence and slot 1's current-turn valence. The main question is whether A_{valence} differs across conditions. The directional-slope LMMs are retained as supplemental views. Starter-mechanism and Human-AI speaker-type decompositions are deferred to a follow-up pass.

1 Load Data

```
## Data availability:

## # A tibble: 6 x 4
##   file                                `human-ai` `human-human` `ai-ai`
##   <chr>                                <lgl>      <lgl>      <lgl>
## 1 dyadic_sentiment_scores.parquet      TRUE      TRUE      TRUE
## 2 interaction_level_surface_metrics.parquet TRUE      TRUE      TRUE
## 3 novelty_scores.csv                   TRUE      TRUE      TRUE
## 4 semantic_exploration_binned.parquet  TRUE      TRUE      TRUE
## 5 story_embeddings_interaction_level.parquet TRUE      TRUE      TRUE
## 6 interaction_level_stories_filtered.csv TRUE      TRUE      TRUE
```

Table 1: Loaded exchange-aligned valence data by condition

condition	n_stories	n_turns	min_turn	max_turn
Human-AI	100	854	1	9
Human-Human	36	324	1	9
AI-AI	40	360	1	9

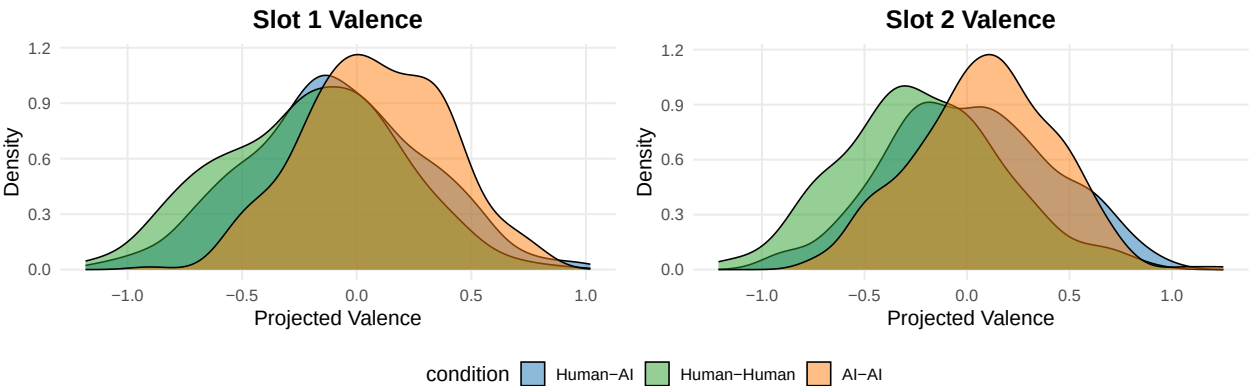
2 Preprocessing

```
## Analysis rows: 2724
## Stories: 176
```

3 Descriptive Checks

Table 2: Mean projected valence by condition

condition	n_stories	n_turns	slot_1_mean	slot_2_mean
Human-AI	100	754	-0.092	0.017
Human-Human	36	288	-0.205	-0.218
AI-AI	40	320	0.080	0.088



4 Primary: Per-Story Asymmetry Magnitude

Stories with both directional correlations available:

Table 3: Per-story A_valence sample sizes by condition

condition	n
Human-AI	100
Human-Human	36
AI-AI	40

Table 4: Per-story |Fisher-z asymmetry| by condition

condition	n	mean_A	sd_A	median_A
Human-AI	100	0.551	0.466	0.436
Human-Human	36	0.463	0.375	0.381
AI-AI	40	0.556	0.920	0.375

One-way ANOVA: A_valence ~ condition

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## condition    2   0.23   0.1160   0.338  0.714
## Residuals  173  59.40   0.3434
```

##

Kruskal-Wallis (robust): A_valence ~ condition

##

Kruskal-Wallis rank sum test

##

data: A_valence by condition

Kruskal-Wallis chi-squared = 2.0009, df = 2, p-value = 0.3677

##

Pairwise Wilcoxon (BH-adjusted):

##

Pairwise comparisons using Wilcoxon rank sum test with continuity correction

##

data: df_asym\$A_valence and df_asym\$condition

##

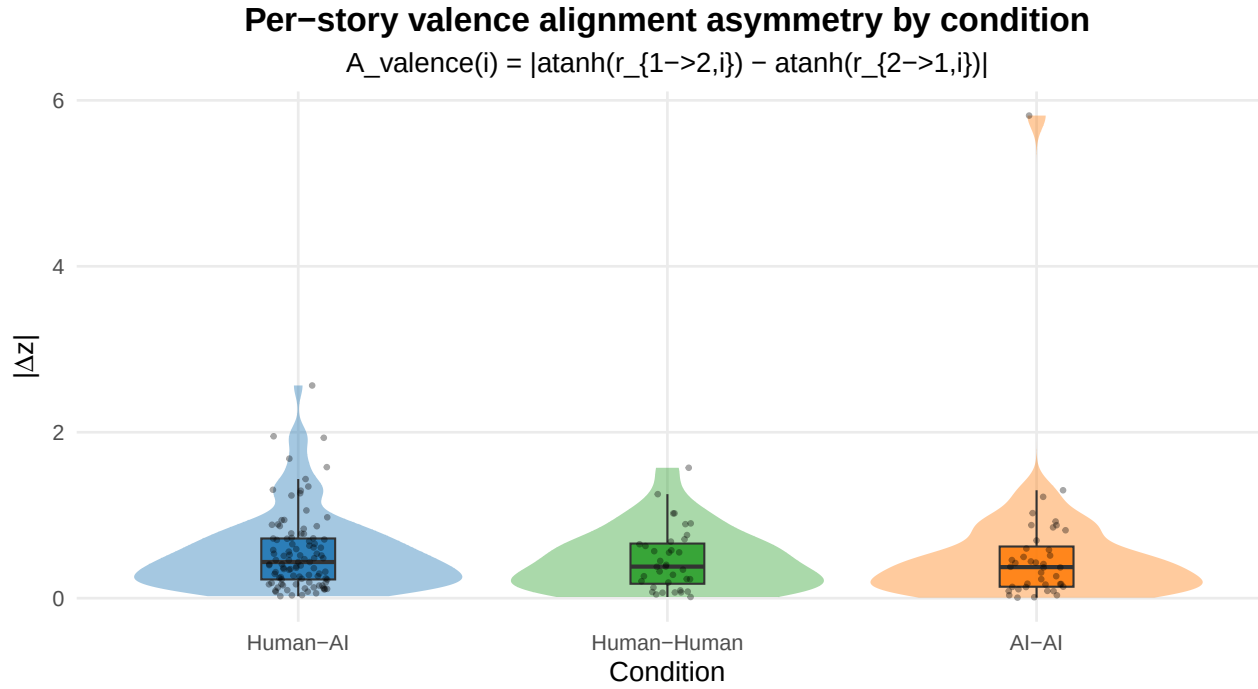
```
##           Human-AI Human-Human
```

```
## Human-Human 0.52      -
```

```
## AI-AI       0.52      0.82
```

##

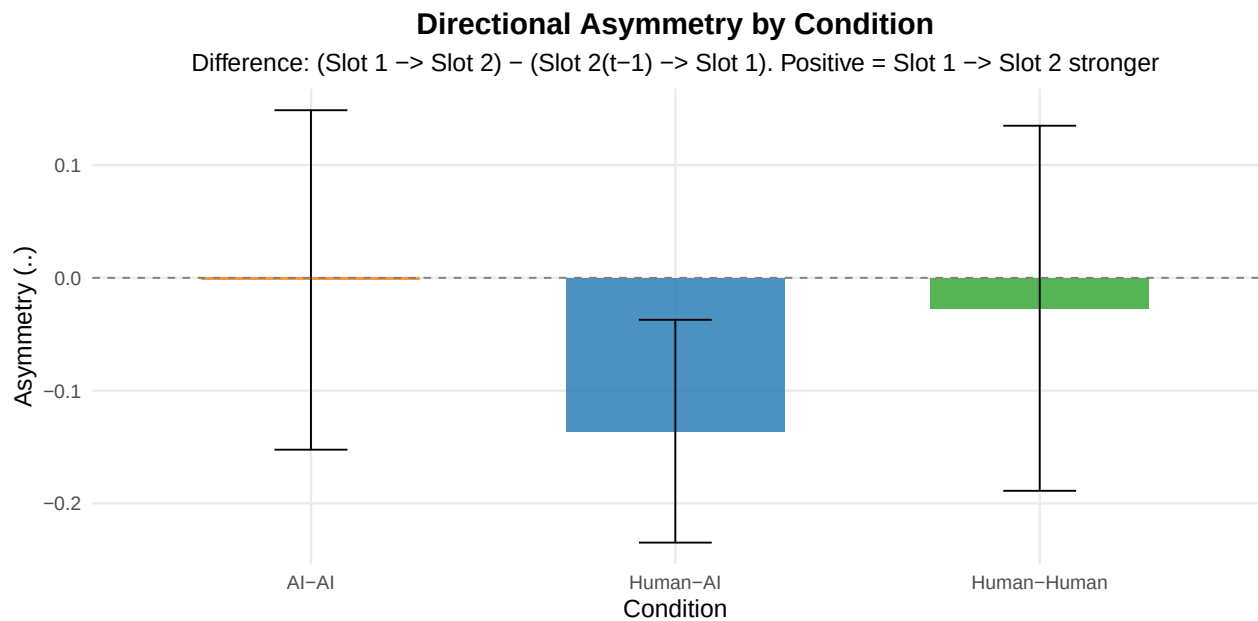
P value adjustment method: BH



5 Supplemental: Within-Condition Directional Asymmetry

Table 5: Within-condition alignment asymmetry by slot direction

condition	slope_slot1_to_slot2	slope_slot2_to_slot1	difference	se	t	p	sig
Human-Human	0.169	0.142	-0.027	0.083	-0.327	0.744	
Human-AI	0.216	0.080	-0.136	0.050	-2.698	0.007	**
AI-AI	0.177	0.176	-0.002	0.077	-0.024	0.980	



6 Supplemental: Cross-Condition Directional Model

```
## Full model summary:

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: response_z ~ predictor_z * alignment_direction * condition +
##          (1 + predictor_z | conversation_id)
## Data: df_std
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: 7341.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.64196 -0.72891 -0.00527  0.72951  2.53254
##
## Random effects:
## Groups          Name          Variance Std.Dev. Corr
## conversation_id (Intercept) 0.00000  0.0000
##                  predictor_z 0.01714  0.1309   NaN
## Residual              0.83615  0.9144
## Number of obs: 2724, groups: conversation_id, 176
##
## Fixed effects:
##
##              Estimate Std. Error    df
## (Intercept)    3.168e-18  1.919e-02 2.542e+03
## predictor_z     1.601e-01  2.326e-02 1.686e+02
## alignment_direction1 -9.256e-18  1.919e-02 2.542e+03
## condition1      -7.729e-18  2.351e-02 2.542e+03
## condition2      -3.149e-18  2.919e-02 2.542e+03
## predictor_z:alignment_direction1  2.748e-02  2.052e-02 2.542e+03
## predictor_z:condition1 -1.175e-02  2.849e-02 1.733e+02
## predictor_z:condition2 -4.740e-03  3.539e-02 1.670e+02
## alignment_direction1:condition1  1.213e-17  2.351e-02 2.542e+03
## alignment_direction1:condition2 -1.021e-17  2.919e-02 2.542e+03
## predictor_z:alignment_direction1:condition1  4.053e-02  2.519e-02 2.542e+03
## predictor_z:alignment_direction1:condition2 -1.399e-02  3.121e-02 2.542e+03
##
##              t value Pr(>|t|)
## (Intercept)    0.000    1.000
## predictor_z     6.882 1.11e-10 ***
## alignment_direction1  0.000    1.000
## condition1      0.000    1.000
## condition2      0.000    1.000
## predictor_z:alignment_direction1  1.339    0.181
## predictor_z:condition1 -0.412    0.681
## predictor_z:condition2 -0.134    0.894
## alignment_direction1:condition1  0.000    1.000
## alignment_direction1:condition2  0.000    1.000
## predictor_z:alignment_direction1:condition1  1.609    0.108
## predictor_z:alignment_direction1:condition2 -0.448    0.654
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
```

```

##          (Intr) prdct_ algn_1 cndtn1 cndtn2 pr_:_1 prd_:1 prd_:2 al_1:1
## predictor_z  0.000
## algnmnt_dr1  0.000  0.000
## condition1  -0.406  0.000  0.000
## condition2   0.207  0.000  0.000 -0.438
## prdctr_z:_1  0.000  0.000  0.000  0.000  0.000
## prdctr_z:c1  0.000 -0.408  0.000  0.000  0.000  0.000
## prdctr_z:c2  0.000  0.207  0.000  0.000  0.000  0.000 -0.438
## algnmnt_1:1  0.000  0.000 -0.406  0.000  0.000  0.000  0.000  0.000
## algnmnt_1:2  0.000  0.000  0.207  0.000  0.000  0.000  0.000  0.000 -0.438
## prdct_:_1:1  0.000  0.000  0.000  0.000  0.000 -0.403  0.000  0.000  0.000
## prdct_:_1:2  0.000  0.000  0.000  0.000  0.000  0.206  0.000  0.000  0.000
##          al_1:2 p_:_1:1
## predictor_z
## algnmnt_dr1
## condition1
## condition2
## prdctr_z:_1
## prdctr_z:c1
## prdctr_z:c2
## algnmnt_1:1
## algnmnt_1:2
## prdct_:_1:1  0.000
## prdct_:_1:2  0.000 -0.439
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

##
## Alignment slopes by condition and direction:

##   condition   alignment_direction   predictor_z.trend      SE  df lower.CL
## Human-AI      Slot 1 → Slot 2           0.2163 0.0381 549  0.14150
## Human-Human    Slot 1 → Slot 2           0.1688 0.0616 496  0.04779
## AI-AI          Slot 1 → Slot 2           0.1775 0.0584 496  0.06267
## Human-AI      Slot 2(t-1) → Slot 1       0.0803 0.0381 549  0.00548
## Human-Human    Slot 2(t-1) → Slot 1       0.1418 0.0616 496  0.02080
## AI-AI          Slot 2(t-1) → Slot 1       0.1756 0.0584 496  0.06079
## upper.CL
##   0.291
##   0.290
##   0.292
##   0.155
##   0.263
##   0.290
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

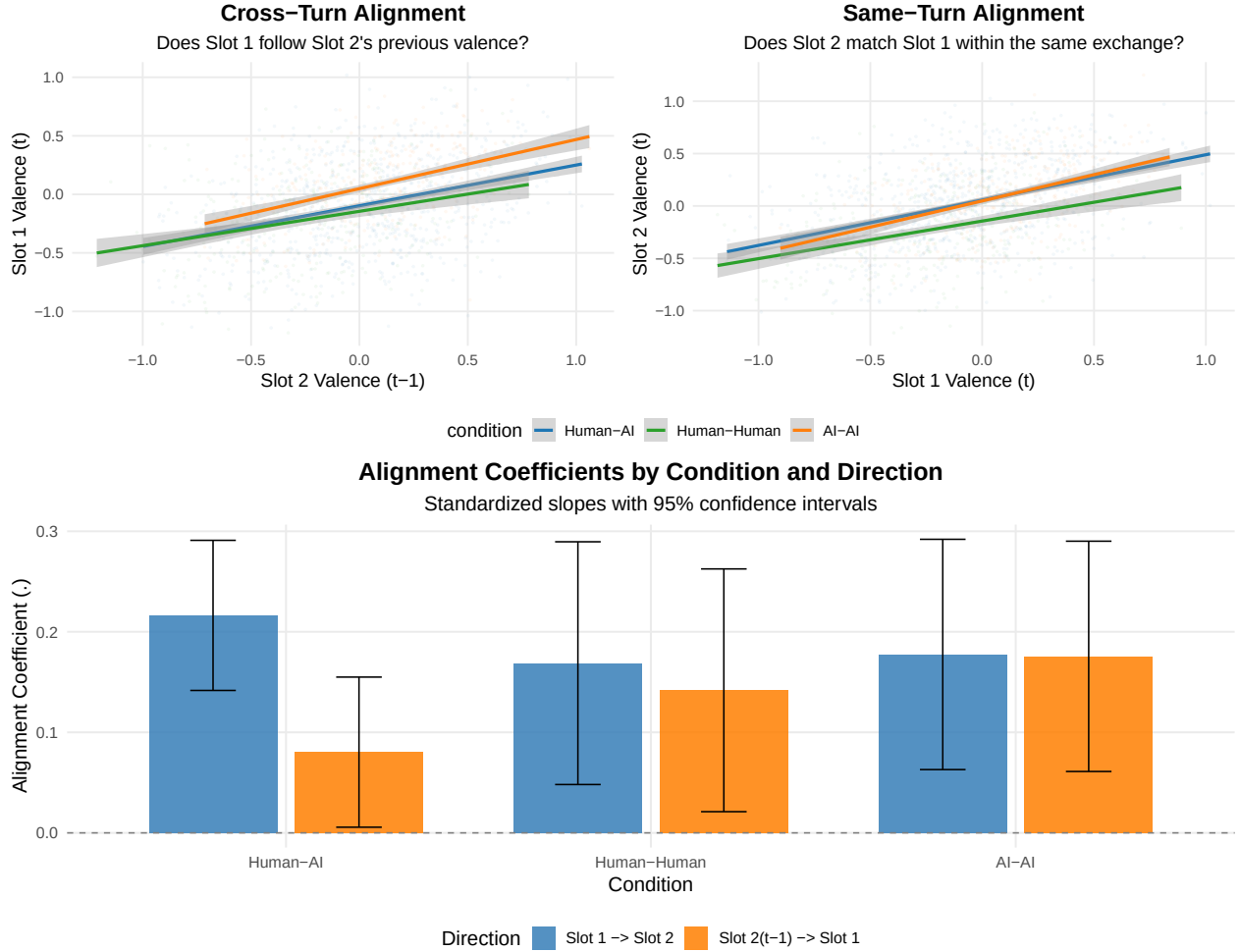
##
## Within-condition direction contrasts:

## condition = Human-AI:
##   contrast                                estimate      SE  df t.ratio p.value
## (Slot 2(t-1) → Slot 1) - Slot 1 → Slot 2 -0.13602 0.0506 2367  -2.690  0.0072
##
## condition = Human-Human:

```

```
## contrast                estimate      SE    df t.ratio p.value
## (Slot 2(t-1) → Slot 1) - Slot 1 → Slot 2 -0.02699 0.0815 2367  -0.331  0.7404
##
## condition = AI-AI:
## contrast                estimate      SE    df t.ratio p.value
## (Slot 2(t-1) → Slot 1) - Slot 1 → Slot 2 -0.00188 0.0773 2367  -0.024  0.9806
##
## Degrees-of-freedom method: kenward-roger
```

7 Supplemental: Directional Scatter Visualizations



8 Summary

Primary cross-condition result: per-story $A_{\text{valence}} = |\text{atanh}(r_{1 \rightarrow 2, i}) - \text{atanh}(r_{2 \rightarrow 1, i})|$, tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon.

Supplemental views retained:

- within-condition directional slope LMMs (two pairing rules)
- full cross-condition LMM with direction 00d7 condition interactions
- directional scatterplots

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)
- Human-AI decomposition (`speaker_type` \u00d7 `speaker_is_starter` within Human-AI only)

Analysis completed: 2026-04-26 20:16:49.003268